
2025 M3C Summary Sheet

SUMMARY

America is a country built on loose change, one that has relied on the back of the country for 2 centuries. However, due to the rise of online technologies and inflation, it has reshaped the use of the penny on the millions of transactions that happen everyday. The goal of this project is to understand the economic impact of removing the penny starting from 2026.

One thing to know is that the pennies do not instantly disappear, but they gather in places such as drawers, jars, glove compartments, cars, and many other places. To find out how long pennies will remain in circulation, it is important to know how many pennies each person has. It uses a Monte Carlo Simulation that categorizes the realistic storage locations and it shows a heavily right-skewed distribution. This means that most individuals hold relatively few pennies, while a few people have loads of pennies, leading us to believe that pennies will still continue to circulate.

As pennies fade from common use, businesses have to decide whether or not to accept them. The project models the penny acceptance as a population-level that is decided by the logistic differential equation we have made. We use the population density and time that has elapsed to reflect on how the rate of companies will not accept the penny anymore. In denser, more urban areas the penny acceptance declines more rapidly as businesses adapt to customer expectations compared to the rural regions who experience a slower change. Over time, the model predicts that pennies will phase out of transactions.

The decline in usage for pennies will lead to companies making a tough decision: how to handle transactions. Currently, transactions are rounded to the nearest cent however without the pennies we have to round to the nearest 5 cents. To address this, the project makes a profit-maximization problem that evaluates the rounding strategies over a fixed time horizon. There are 3 major forces, the expected rounding gains on cash transactions, avoided credit-card processing fees, and the customer behavior response. Customers are assumed to not leave the business entirely but they switch from cash to card when it is unfair as perceived by them.

Behavior sensitivity is modeled as a function of rounding magnitude relative to item price and capturing the intuition that the five-cent difference matters far more on a low-cost purchase than an expensive one. This causes the cash usage to decay at an exponential rate, especially as rounding becomes more aggressive. The resulting profit equation will allow the users to determine the optimal rounding bias which is able to balance the gains against the long-term behavior.

The model shows that the aggressive rounding down is optimal for much lower priced items where the behavioral effects are almost negligible. However, as the item price increases of the rounding unit become larger (from 5 cents to 10 or 25), the customer sensitivity is more valued and the neutral rounding strategies become more optimal. These results show that rounding policy is not merely a pricing decision, but a strategic choice that influences payment-method composition across time.

Together, these models are able to show the effects of the pennies being removed from the government. Coins accumulate, businesses adapt, and consumers change responses and by integrating together stochastic simulation, population dynamics, and profit optimization we are able to solve for the insights that businesses have and how policy makers can effectively work in a penny-free environment. Because when every cent counts, understanding how people respond to losing one matters more than the coin itself.

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1 INTRODUCTION

1.1 Background

With the advent of the credit card, the use of physical currency has been on the decline for many years. In 2018, only 18% of transactions used cash, compared to 31% just 2 years before [3]. Thus, modeling the transition of society into a cashless system is necessary to predict future social and fiscal outcomes. The logical first step in moving towards a cashless society would be to phase out lower monetary denominations.

Since its inception in 1793, the penny has remained the lowest monetary denomination in the United States. Though it has remained a familiar part of everyday transactions, rising prices and changes in payment habits have led to a fundamental change in its daily usage. In fact, many countries, such as Canada, Australia, New Zealand, and the U.S. soon to follow, have eliminated their lowest denomination all together [5].

Currently, the U.S. penny remains in circulation, though its future is uncertain. Everyday transactions often involve change, but the general public do not spend change as often as they receive it. This puts additional pressure on the U.S. Mint to continuously mint new pennies at production costs more than their face value [11]. More than 3.2 billion pennies were minted in 2024 alone, with anywhere from 114 to 240 billion pennies estimated to be in circulation [10, 4].

The U.S. Mint has officially announced the discontinuation of penny minting in 2026, leading to a variety of interesting and important questions that we address in this paper [2].

1.2 Problem Interpretation

1. Determine the amount of pennies in the possession of the average American.
2. Predict the amount of pennies in circulation after the U.S. Mint discontinues penny production.
3. Forecast the percentage of American businesses that will accept pennies x years after 2026.
4. Model the optimal rounding strategy for transactions not divisible by five cents.

1.3 Deliverables

1. Quantitative model for determining pennies in the possession of the average American.
2. Quantitative regression model for the future percentage of businesses that accept pennies as payment.
3. Quantitative rounding model for the optimal business strategy post-Mint cessation.
4. Modified rounding model for post-nickel-and-dime-Mint cessation.

1.4 Plan of Action

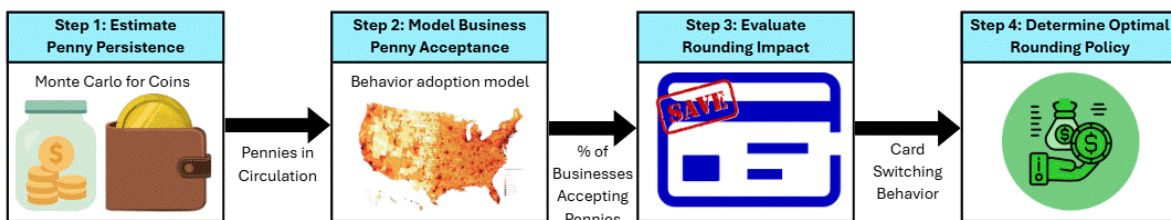


Figure 1: Flowchart of action plan

The project aims to evaluate the economic and behavioral impacts of eliminating the penny from circulation. This is accomplished by first estimating the number of pennies remaining in the possession of the average American using a Monte Carlo simulation of realistic storage locations. Simultaneously, the project models how

businesses gradually stop accepting pennies over time based on population density and behavioral adoption. Using the results of these models, a profit optimization framework is applied to determine optimal rounding strategies once pennies are no longer accepted, accounting for rounding gains, avoided credit card fees, and consumer payment switching. By integrating these components, the project identifies rounding policies that maximize long-term profitability while minimizing adverse behavioral effects.

2 PENNIES OF THE AVERAGE AMERICAN

This model involves iterative random sampling to predict the amount of pennies that an average American owns. This is done through a Monte Carlo simulation, which is used when there exists uncertainty in certain input variables. In this problem, the probability of different people having pennies in various locations changes, so a Monte Carlo simulation is appropriate.

We divided an individual's ownership of pennies into five components: their wallet, car, a dedicated penny jar, drawers, and miscellaneous locations. These components follow different statistical distributions, which is reflected in our model.

2.1 Assumptions and Justifications

2.1.1 Equations

The first component of penny ownership is within an individual's wallet. The statistical distribution chosen for this component was a Poisson distribution, which describes the probability of a given number of events occurring in a fixed interval of time if these events occur with a known constant mean rate and independently of the time since the last event. The sole input parameter for the Poisson distribution is λ , which represents the average rate or expected number of events in a specific time interval.

The probability mass function, or PMF, of the Poisson distribution is given as follows

$$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}, \quad k = 0, 1, 2, \dots \quad (1)$$

For this component of Model I, $\lambda = 6.92$, as derived from [7]. This means that the average amount of pennies in a wallet is 6.92.

The next main source of penny ownership is in their cars. Since the distribution of pennies within cars differs fundamentally from that of the distribution within wallets, a different statistical distribution is required.

This component of Model I involved the combination of a Bernoulli distribution and a Negative Binomial distribution. The justification of the distribution choices is that a Bernoulli distribution is used when there is a boolean distinction between a condition. In this case, whether an individual has pennies in their car. The Negative Binomial distribution models the number of Bernoulli trials needed for a fixed number of successes, meaning a combination of these distributions results in a comprehensive evaluation of penny ownership in cars across a population.

The input parameter for the Bernoulli distribution is p , which is the probability of the desired condition X being met, where $p = P(X = 1)$. X is a Bernoulli random variable that represents a boolean state of success or failure. For this model, $p = 0.88$ as derived from [5, 1].

The PMF of the Bernoulli distribution follows

$$P(X = x) = p^x (1 - p)^{1-x}, \quad x \in \{0, 1\} \quad (2)$$

The Negative Binomial distribution takes two parameters. The Negative Binomial random variable X is defined as the number of failures before the r -th success, where r represents the target number of successes. Similar to the Bernoulli distribution, p represents the probability of attaining a successful trial, or X . For this model, $p = 0.57143$ and $r = 2.6667$. The mean and variance, respectively, are given by the following equations

$$\text{Mean: } \mu = \frac{r(1-p)}{p}, \quad \text{Variance: } \sigma^2 = \frac{r(1-p)}{p^2} \quad (3)$$

With the given values of r and p , $\mu = 2$ and $\sigma^2 = 3.5$. The PMF of the Negative Binomial distribution is given by

$$P(X = k) = \binom{k+r-1}{r-1} p^r (1-p)^k, \quad k = 0, 1, 2, \dots \quad (4)$$

The third component of Model I is pennies found in dedicated penny jars. Once again, the distribution of this component is fundamentally different than the previous two, hence the use of a modified Lognormal distribution. The Lognormal distribution is used to model positive, right-skewed data of a random variable X . The Lognormal distribution is a modification to the Normal Distribution and is used if the natural logarithm of X follows a normal distribution, or

$$\ln(X) \sim \mathcal{N}(\mu, \sigma^2) \quad (5)$$

The Lognormal distribution takes two parameters: μ , the mean of $\ln(X)$, and σ , the standard deviation of $\ln(X)$. For the model, $\mu = 4.3$ and $\sigma = 1.2$ as derived from [9, 6]. However, since the domain of X is $x \in (0, \infty)$, the distribution must be modified for this model, as it is infeasible for X to exceed a certain capacity. We set the maximum attainable X to be 2000 as a reasonable ceiling. The Lognormal distribution does not have a PMF, as it is a continuous, rather than discrete, distribution. Instead, it has a probability density function, or PDF. The PDF of the Lognormal distribution is given by

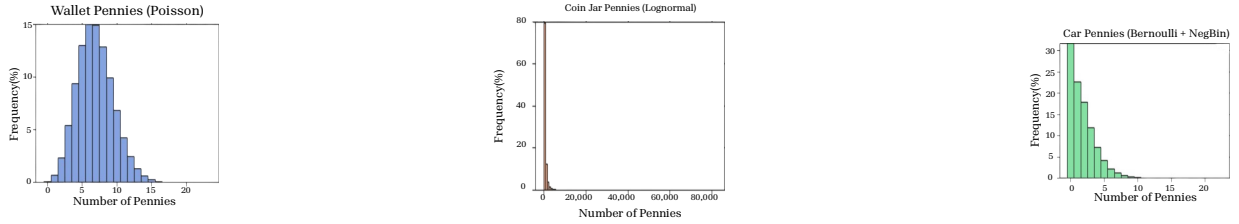
$$f_X(x) = \frac{1}{x\sigma\sqrt{2\pi}} \exp\left(-\frac{(\ln x - \mu)^2}{2\sigma^2}\right), \quad x > 0 \quad (6)$$

The two remaining components, the penny drawer and miscellaneous, both follow similar distributions and can be expressed using a combination of the aforementioned Bernoulli and Poisson distributions.

The penny drawer takes $p = 0.95$ for the Bernoulli component and $\lambda = 0.3$ for the Poisson component, while the miscellaneous pennies take $p = 0.75$ for the Bernoulli component and $\lambda = 0.15$ for the Poisson component [8].

2.2 Results of Model I

Simulating each model, we get these distributions with a million sample Monte Carlo Simulation:



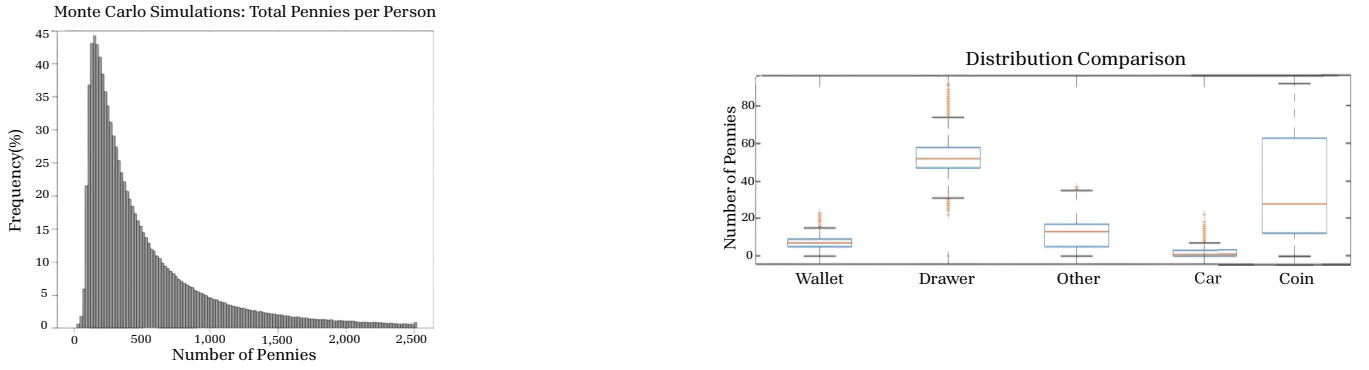
(a) The estimated frequency of a certain number of pennies in someone's wallet (b) The estimated frequency of a certain number of pennies in someone's jar (c) The estimated frequency of a certain number of pennies in someone's car



(d) The estimated frequency of a certain number of pennies in someone's drawers (e) The estimated frequency of a certain number of pennies in miscellaneous locations

Figure 2: The estimated frequency of certain penny quantities in various locations

Combining these, we get this distribution:



(a) Histogram of Americans and how many pennies they have.

(b) A box and whiskers plot of the 5 distribution. Note that the coin value is capped at 100 because technically the distribution is infinite

Figure 3: The estimated frequency of certain penny quantities in various locations

Statistical analysis of this distribution in MATLAB yielded the following values:

$$\begin{aligned}
 \mu &= 695.63 & 95\% \text{ CI} &= [97, 3261] \\
 \sigma &= 1115.20 & 99\% \text{ CI} &= [72, 6740] \\
 \text{Med} &= 375.54 & 99.9\% \text{ CI} &= [38, 15651] \\
 \text{IQR} &= 549
 \end{aligned} \tag{7}$$

2.3 Conclusion

The results show an average of 695.63 pennies per person, for an estimated value of \$6.96. The median of the data shows a strong right skew, which agrees with the initial hypothesis and follows the expected Negative Binomial distribution. Additionally, the IQR shows a large range of values in the middle 50% of the data, showing the spread of values even in typical individuals. The confidence intervals of the data show that 99% of the data lie between 72 and 6740 and 99.9% of the data lie between 38 and 15651, which also indicates the large spread of data, as expected.

3 BUSINESS PENNY ACCEPTANCE POLICY

This model predicts the percentage of cash-accepting businesses that will continue to accept pennies in the years following 2026, when the U.S. Mint ceases production of new pennies. Because the decision to accept or reject pennies evolves over time and depends on both economic costs and collective behavior, we model penny acceptance as a dynamic process rather than a static estimate. A continuous-time decay model is used to capture how acceptance decreases as pennies become less common in circulation and increasingly inconvenient to handle.

We represent penny acceptance using a single state variable: the fraction of cash-accepting businesses that still accept pennies at a given time. The rate at which businesses stop accepting pennies depends on two main factors: baseline operational inconvenience and social feedback effects, where declining penny usage by customers further accelerates rejection by businesses. These interacting effects are incorporated into a nonlinear differential equation, allowing the model to reflect the slow initial decline and faster phase-out observed in similar currency eliminations.

3.1 Assumptions and Justifications

#	Assumption	Justification
1	Statistical homogeneity at a fixed density d .	The model considers the aggregate probability function $P(d, t)$ rather than an individual-based approach. Therefore, any business-specific information is not relevant to the model results, making statistical homogeneity reasonable at a fixed density d .
2	Penny acceptance policies are irreversible.	Once businesses adopt or reject penny acceptance, it is assumed they do not reverse this decision immediately. This reflects real-world behavioral inertia.
3	Decay is instantaneously proportional to current acceptance level P .	The decay rate of acceptance is assumed proportional to the current level, reflecting common logistic-type processes in population dynamics.
4	Decay saturation is based on surrounding population dynamics.	As acceptance declines, the rate of change slows due to saturation effects influenced by the surrounding business population.
5	The interactions between assumptions 4 & 5 results in logistic behavior P .	Combining proportional decay and saturation naturally produces logistic dynamics, consistent with observed adoption/decay trends.
6	A higher population density d results in rapid decay of P .	Denser business environments accelerate peer-influenced decay in acceptance levels, reflecting social influence and competition effects.
7	Spatial independence exists between different densities d_i .	Interactions between businesses at different densities are negligible, simplifying the model to a set of independent logistic processes.
8	There exists a deterministic elimination of fluctuations as $t \rightarrow \infty$.	Over long timescales, stochastic variations average out, allowing the system to converge to a deterministic equilibrium.

3.1.1 Variables

Variable	Significance
$P(d, t)$	The proportion of businesses that accept pennies as payment, as a function of population density d and time t .
d	Population density of a region, in people per square mile (people/mi ²).
t	Time in years (or relevant time units) over which penny acceptance evolves.
$k(d)$	Decay rate parameter, which depends on population density d and controls the rate at which acceptance declines.
a	Logistic model parameter controlling the initial slope of decay.
$C(d)$	Carrying capacity of penny acceptance for density d , representing the minimum acceptance level as $t \rightarrow \infty$.
P_0	Initial proportion of businesses accepting pennies at $t = 0$.

3.1.2 Equations

Logistic Ordinary Differential Equations (ODEs) are widely used to model population dynamics in a controlled, realistic environments. For this problem, a Logistic ODE successfully encapsulates the main

constraints, such as time and population density. Therefore, a Logistic ODE modeled with respect to these parameters, along with an initial population density sensitivity would provide an accurate proportion of American businesses that would accept the penny t years after its discontinuation in 2026.

The model's equation uses the general form of the reaction-diffusion ODE. This equation is given by

$$\frac{\partial P}{\partial t}(d, t) = D \frac{\partial^2 P}{\partial d^2}(d, t) - k(d) P(d, t)(1 - P(d, t)) \quad (8)$$

The justification of a diffusion model is that the change in the proportion of businesses accepting pennies as payment, or D , is influenced by the payment dynamics of surrounding businesses and areas, given the population density and density-dependent decay rate. In the equation, D is the diffusion coefficient, $k(d)$ is the density-dependent decay rate, d is the population density, and t is the time in years after 2026.

If $D = 0$, the general form reduces into a system of logistic first-order ODEs where the principal equation with a fixed d is given by

$$\frac{\partial P(d, t)}{\partial t} = -k(d) P(d, t)(1 - P(d, t)) \quad (9)$$

To model real-world dynamics and logistic regression, the optimal decay rate is exponential. Therefore, $k(d)$ is given by

$$k(d) = a \ln(d + 1) \quad (10)$$

If $a > 0$, the model exhibits steeper decay in high-population-density areas, which is accurate as the rate of decrease in transactions involving pennies will increase in areas with higher population densities.

Using Separation of Variables, this ODE becomes

$$\ln\left(\frac{P(d, t)}{1 - P(d, t)}\right) = -k(d) t + C(d), \quad P(d, t) = \frac{1}{1 + C(d) e^{k(d)t}} \quad (11)$$

In this form, $C(d)$ is given by

$$C(d) = \frac{1 - P_0}{P_0} \quad (12)$$

In the problem, $P_0 = 0.99$, as approximately 99% of businesses will continue to accept pennies as soon as future production is discontinued. The value of a is determined by fitting this equation to the benchmark values provided by the datasets. Given the following inputs of d and t , $P(d, t)$ outputs as follows.

$$\begin{aligned} P(d = 3000, t = 5) &\approx 0.15 \\ P(d = 100, t = 5) &\approx 0.70 \\ P(d, 0) &= 0.99 \end{aligned} \quad (13)$$

All of these benchmarks independently yield $a = 0.1603$ when fitted to the above ODE derived by Separation of Variables.

3.2 Results of Logistic Model

Given $P_0 = 0.99$ and $a = 0.1603$, we construct the following table of values for $k(d)$ and $P(d, t)$ over the time period $t \in [0, 10]$.

Table 1 confirms the initial hypothesis as an increase in d correlated to rapid declines in $P(d, t)$ over time, thereby cohesively connecting with the proposed logistic ODE.

Density d (people/mi ²)	Classification	$k(d)$	$P(d, 0)$	$P(d, 2)$	$P(d, 5)$	$P(d, 10)$
100	high-density rural	0.738	0.99	0.958	0.712	0.0580
500	low-density suburban	0.995	0.99	0.931	0.407	0.0050
1600	mid-density suburban	1.181	0.99	0.903	0.213	0.0010
3000	high-density suburban	1.281	0.99	0.884	0.141	0.0004
10000	low-density urban	1.474	0.99	0.839	0.059	≈ 0

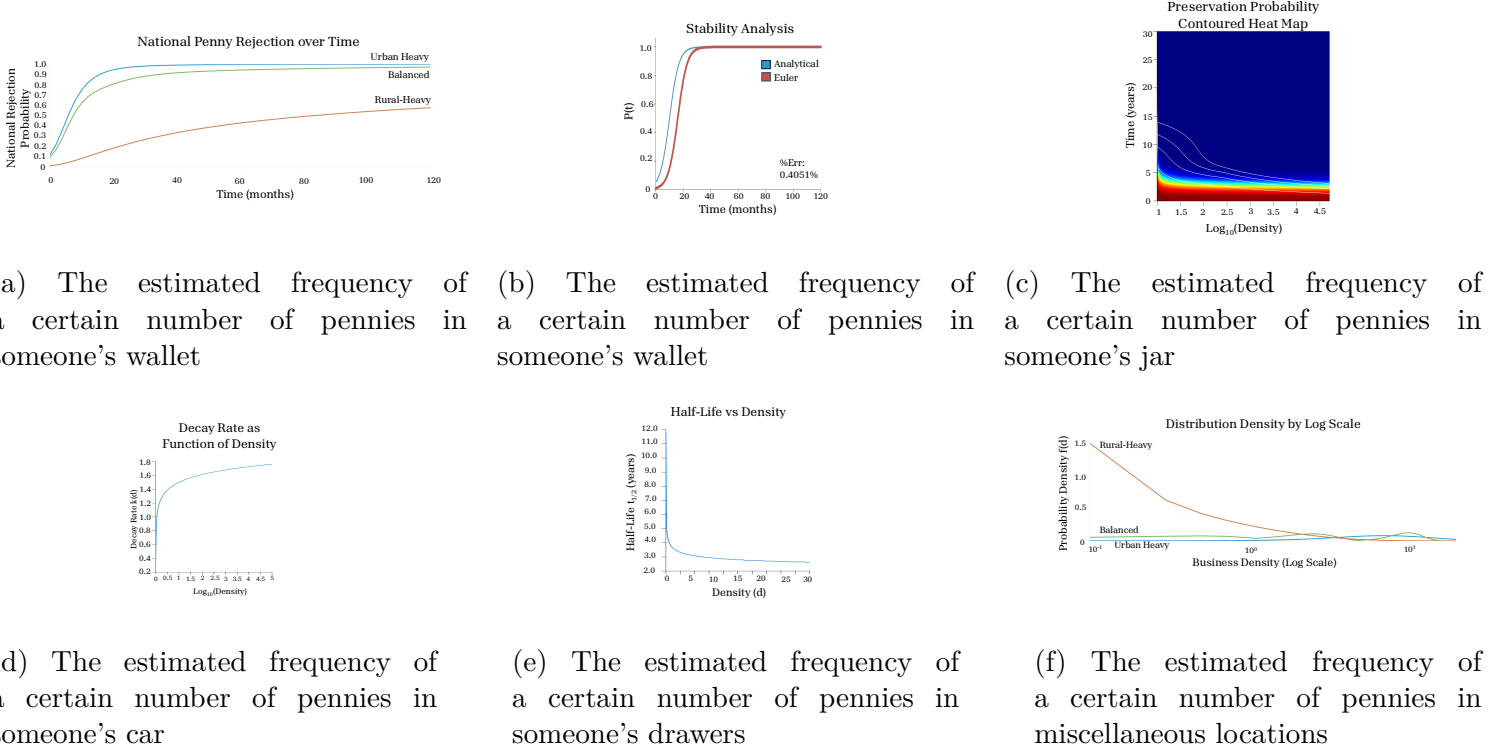
Table 1: $k(d)$ and $P(d, t)$ values for varying population densities d .

Figure 4: The estimated frequency of certain penny quantities in various locations

3.3 Conclusion

This model predicts the future percentage of American businesses that will accept pennies as payment through the use of a logistic Ordinary Differential Equation dependent on a region's population density d and the amount of time elapsed after the penny's discontinuation by the U.S. Mint in 2026 in years t .

4 BUSINESS ROUNDING STRATEGY

This model finds the optimal (profit-maximizing) method of rounding payments after pennies are no longer accepted. To do this, it integrates profit over time, considering factors such as credit card fees and loss of cash-paying customers due to unfair rounding schemes.

4.1 Assumptions and Justifications

Assumption	Justification
Customers do not leave the business due to rounding	Empirical evidence suggests consumers switch payment methods rather than abandoning businesses when faced with rounding.
Baseline cash usage remains constant in the absence of rounding bias	Penny acceptance does not change intrinsic preferences for cash; only perceived unfair rounding affects behavior.
Cash-to-card switching occurs gradually over time	Behavioral adoption and abandonment processes are well-approximated by exponential decay models.
Customers respond to rounding magnitude relative to item price	A fixed rounding amount is more noticeable on low-priced items than on high-priced items.
Credit card processing fees are invisible to customers	Processing fees are paid by businesses and do not directly influence customer payment decisions.
Transaction remainders are uniformly distributed	Retail prices modulo the rounding unit are approximately uniform in real-world transactions.
Profit is evaluated over a fixed planning horizon	Businesses choose rounding policies over discrete time periods rather than continuously updating strategies.

Table 2: Model III Assumptions and Justifications

4.2 Application of Model III

4.2.1 Variables

Variable	Description
θ	Rounding bias ($\theta = 0$ corresponds to neutral rounding, $\theta = 1$ corresponds to always rounding up).
r	Rounding unit (e.g., \$0.05, \$0.10, \$0.25).
p	Price of the item or transaction amount.
C_0	Baseline fraction of customers paying with cash in the absence of rounding bias.
α	Credit card processing fee rate as a fraction of the transaction price.
k	Behavioral sensitivity constant governing cash-to-card switching.
λ	Cash-to-card switching rate induced by rounding.
T	Fixed evaluation horizon for profit calculation.
t	Time variable.
c	Credit card fee rate for companies.

Table 3: Model 3 Variables

4.2.2 Equations

This model involves abstraction into three distinct components: behavioral sensitivity, a cash-paying fraction as a function of time, and a profit per cash transaction function. Individually, these components represent the behavioral decay rate as the penny is phased out, the change in the amount of cash transactions over time, the microeconomic payoff on a per-transaction basis, and overall behavioral and economic factors.

The first component of the model is the behavioral decay rate, or sensitivity. This is given by λ , which represents the responsiveness of customers to the changes in rounding policy based on policy change aggressiveness.

The second key component of the model is the cash-paying function with respect to time, $C(t)$, which represents the fraction of transactions paid in cash at time t . $C(t)$ is given by

$$C(t) = C_0 \exp(-k \frac{r\theta}{p} t) \quad (14)$$

The final aspect of the model is the profit per cash transaction function, which is given by

$$\frac{r\theta}{2} + \alpha p \quad (15)$$

where cp represents the credit card fee for a product of price p .

Together, these components form the final model, which can be expressed as an integral with respect to time t , given as follows

$$J(\theta, p) = \int_0^T C(t) (\frac{r\theta}{2} + \alpha p) dt \quad (16)$$

In its full form, $J(\theta, p)$ is given by

$$J(\theta, p) = \int_0^T C_0 \exp(-\lambda r \theta t) (\frac{r\theta}{2} + \alpha p) dt \quad (17)$$

where the bounds of the integral are the evaluation horizon until the current year.

Evaluating the integral form of $J(\theta, p)$ gives a function form of J in terms of θ and p .

$$J(\theta, p) = C_0 \left(\frac{r\theta}{2} + \alpha p \right) \frac{1 - e^{-\lambda r \theta T}}{\lambda r \theta} \quad (18)$$

4.3 Results of Model III

Maximizing the value of J given p , gives the optimal value of θ . Using a credit card fee of $\alpha = 0.02$, which is standard, a λ of 0.8, and a r of 0.05, since that is what we are rounding to, we can fill in the profit section of the equation. In addition, using a time period of $T = 250$ day, we get a model that can be graphed.

A graph below shows profits for different values of p and θ :

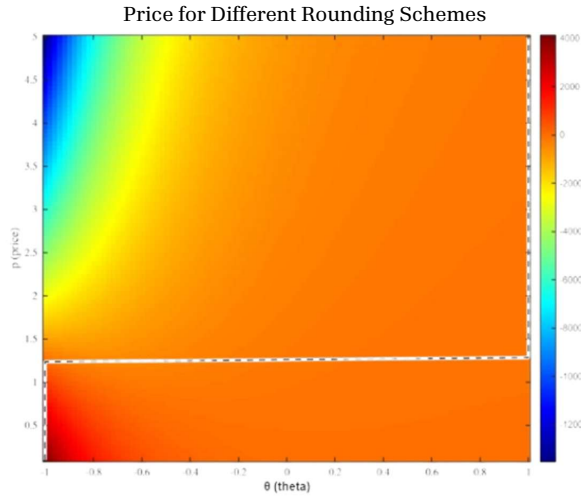
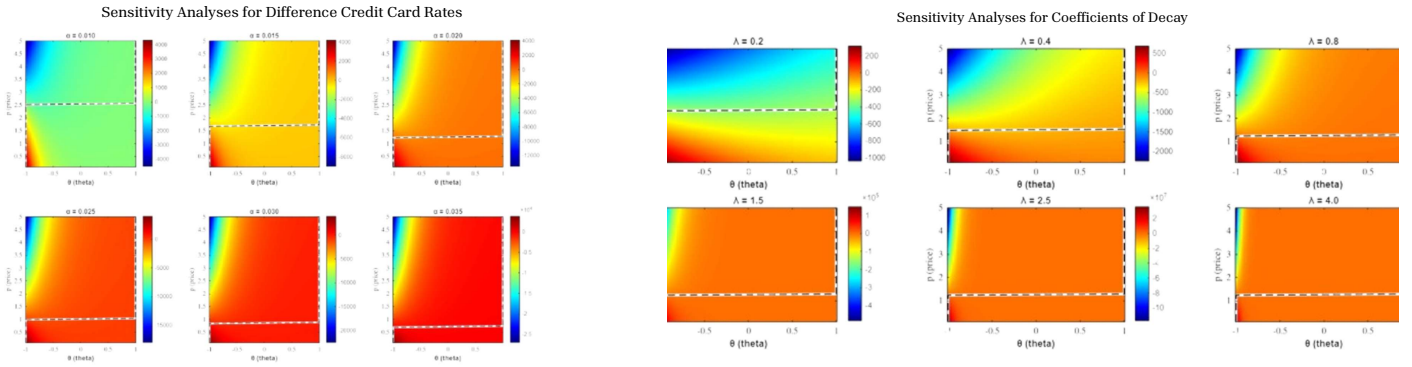


Figure 5: The estimated frequency of a certain number of pennies in someone's wallet

For $p < 1.25$, $\theta = -1$ is optimal and for $p > 1.25$, $\theta = 1$ is optimal. Since $p = 1.25$ represents the shift in optimal θ value, we recommend that businesses round down the prices when paid in cash for transaction totals less than \$1.25. For transaction totals above \$1.25, we recommend businesses round up payment totals.

In order to test the robustness of this model, a sensitivity analysis was performed on the α , the rate the credit card companies charge. Another analysis was performed on λ which is how reactive customers are to the rounding scheme. The results are shown below:



(a) Sensitivity analysis was performed on the α , the rate the credit card companies charge. (b) Sensitivity analysis performed on λ which is how reactive customers are to the rounding scheme

Figure 6: The estimated frequency of certain penny quantities in various locations.

In order to see how these two sensitivities interact, they were combined onto a plot where p and θ were held constant:

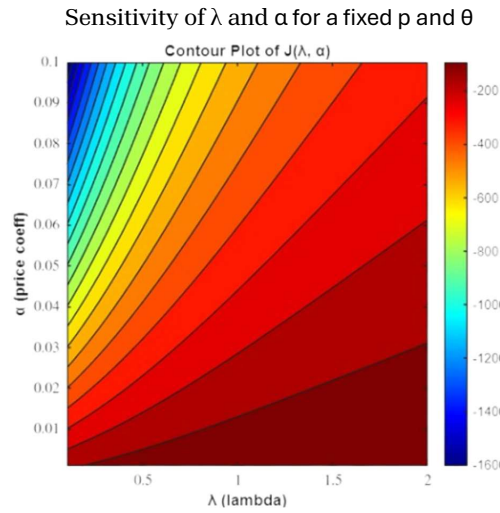


Figure 7: Combined 2D Sensitivity Analysis

Finally, these two graphs show the effect of λ and α on the optimal θ to switch from a rounding down to a rounding up strategy.

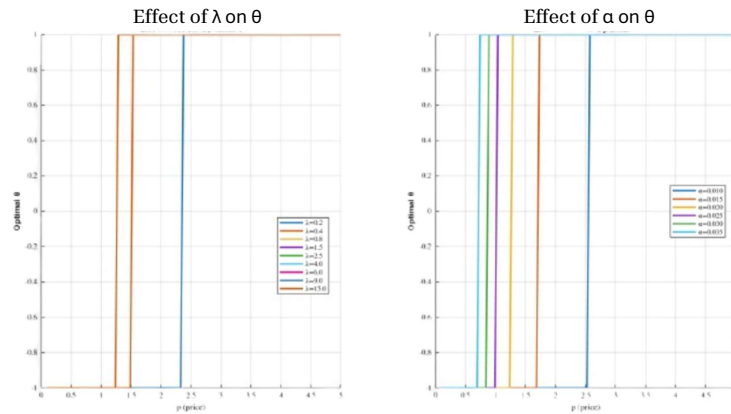


Figure 8: Combined 2D Sensitivity Analysis

4.4 Conclusion

This model predicts the optimal rounding strategy for businesses after the U.S. Mint discontinues the production of new pennies in 2026. Using an exponential decay model, paired with an integral to describe the accumulation of profits over time, we determined that the optimal cutoff for rounding up or down transaction totals occurs at \$1.25.

5 MODEL DISCUSSION

5.1 Strengths

1. The model derives the decline in cash usage from accumulated customer dissatisfaction rather than assuming an exogenous decay function.
2. Cash abandonment emerges naturally from the coupled differential equations, strengthening the interpretation of the results.
3. Each parameter has a clear real-world meaning, enabling meaningful sensitivity analysis.
4. The framework extends naturally to the removal of nickels, dimes, or quarters by adjusting the rounding unit without altering the core structure.
5. The model does not rely on memoryless or constant hazard assumptions, or random variable that were added to make the model make more sense.

5.2 Weaknesses

1. The system of differential equations generally lacks a closed-form solution, requiring numerical methods.
2. Introducing behavioral parameters increases flexibility but also raises challenges in justification and estimation.
3. The model assumes uniform customer responses, neglecting heterogeneity in preferences and payment behavior, especially in Model 3, which doesn't consider density as a variable at all.

6 CONCLUSION

The goal of this project was to understand the economic effect of stopping to mint pennies. More specifically, the goal was to analyse how long pennies last, how would business react, and which rounding strategies would maximize the profit. The elimination of pennies is vast as business would part and change prices, pennies would phase out, and would create a cascade effect in how people use pennies.

For model one, the results show that the average amount of pennies that an American has is 700 pennies, a heavily right skewed value, with a majority of them being stored in jars, drawers, cars, and other locations. Additionally, pennies will last for years and the Monte Carlo showed that the physical persistence of those pennies would not match the economic relevance of them. In model two, the business stops accepting pennies at a much faster rate than with which they become obsolete and the population density of towns only accelerates this change. Many businesses will adopt to remove the penny and this made it crucial for our 3rd model to figure out how to effectively round to the nearest 5 cents to maximize profit. In model 3, the main takeaway was that the aggressive rounding path would maximize the immediate rounding revenue but it also drives the shift from cash to cards. By making it so that the rounding method changed based on the price of the item, it would be able to minimize the backlash and the profit. For lower price items, ones under 1.25 dollars, it is better to round the price down while for any purchase above that price, it is more cost effective to round the price up.

Using these models, some inter-model connections can be made such as how the penny persistence in model one shows that the physical coins stay around but the acceptance of the new type of monetary system is adapted to much faster. Also, since the decline in acceptance in model two pairs with the rounding strategy in model 3, it shows how these models connect together to make a deeper message.

Something that business should do is prepare for the rounding issues and the influences it will have on entire payment ecosystems and card-fee economics. Additionally, removing coins in the future would also result in very similar values and help provide a framework for a cashless society. Also, the gap between physical availability and economic acceptance reveals how quickly norms evolve. Some future directions to make this project better is to introduce agent based learning or machine learning. Also, it would be beneficial to add inflation and dynamic pricing and model how this affects low-income people. In conclusion, understanding behavioral responses is essential for designing fair, profitable, and sustainable rounding policies.

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8 APPENDICES

8.1 Appendix 1: Code for Monte Carlo Simulation

`./model1.py`

```

1 % Monte Carlo Simulation of Penny Distribution
2 % Based on research-backed parameters for different storage locations
3 % Number of simulations: 1,000,000 per category
4
5 clear; clc;
6
7 % Set random seed for reproducibility
8 rng(42);
9
10 % Number of simulations
11 n = 1000000;
12
13 %% 1. WALLET - Poisson( = 6.92)
14 % Source: Pudwell & Rowland "What's in Your Wallet?"
15 lambda_wallet = 6.92;
16 wallet_pennies = poissrnd(lambda_wallet, n, 1);
17
18 %% 2. COIN JAR - Lognormal( = 4.3, = 1.2)
19 % Source: U.S. household coin jar data (median $60-$90)
20 % Convert dollars to pennies: median $75 = 7500 pennies
21 mu_jar = 5.717; % ln(median in pennies)
22 sigma_jar = 1.2;
23 coinjar_pennies = lognrnd(mu_jar, sigma_jar, n, 1);
24
25 %% 3. CAR - Bernoulli + Negative Binomial
26 % Source: Car spare change survey (88% had coins)
27 % Step 1: Bernoulli for presence of any cash
28 p_car_presence = 0.88;
29 car_has_cash = rand(n, 1) < p_car_presence;
30
31 % Step 2: If cash present, negative binomial for number of pennies
32 % Target: mean | 2, variance | 3-4
33 mean_car_pennies = 2;
34 var_car_pennies = 3.5;
35
36 % Convert to negative binomial parameters
37 % mean = r(1-p)/p, variance = r(1-p)/p^2
38 % Solving: r = mean^2 / (var - mean)
39 r_car = mean_car_pennies^2 / (var_car_pennies - mean_car_pennies)
40 p_car_nb = mean_car_pennies / var_car_pennies
41
42
43 car_pennies = zeros(n, 1);
44 car_pennies(car_has_cash) = nbinrnd(r_car, p_car_nb, sum(car_has_cash), 1);
45
46 %% 4. DRAWER - Bernoulli + Poisson
47 % Source: U.S. household coin accumulation patterns
48 % Step 1: Bernoulli for whether drawer contains any cash
49 p_drawer_presence = 0.95;
50 drawer_has_cash = rand(n, 1) < p_drawer_presence;
51
52 % Step 2: If cash present, Poisson for number of pennies
53 lambda_drawer = 53;
54
55 drawer_pennies = zeros(n, 1);
56 drawer_pennies(drawer_has_cash) = poissrnd(lambda_drawer, sum(drawer_has_cash), 1);
57
58 %% 5. OTHER - Bernoulli(p=0.75) + Poisson(=15)
59 % Source: Surveys on spare change presence
60 p_presence = 0.75;
61 lambda_other = 15;
62
63 % Bernoulli trial for presence
64 presence = rand(n, 1) < p_presence;
65
66 % Poisson count given presence
67 other_pennies = zeros(n, 1);
68 other_pennies(presence) = poissrnd(lambda_other, sum(presence), 1);
69
70 %% SUMMARY STATISTICS
71 fprintf('=== PENNY DISTRIBUTION SIMULATION RESULTS ===\n\n');
72
73 fprintf('WALLET (Poisson=6.92):\n');
74 fprintf(' Mean: %.2f pennies\n', mean(wallet_pennies));
75 fprintf(' Std Dev: %.2f\n', std(wallet_pennies));
76 fprintf(' Median: %.2f\n', median(wallet_pennies));
77 fprintf(' Range: [%d, %d]\n', min(wallet_pennies), max(wallet_pennies));
78
79 fprintf('COIN JAR (Lognormal =%.2f, =%.2f):\n', mu_jar, sigma_jar);
80 fprintf(' Mean: %.2f pennies ($%.2f)\n', mean(coinjar_pennies),
81 mean(coinjar_pennies)/100);
82 fprintf(' Std Dev: %.2f\n', std(coinjar_pennies));
83 fprintf(' Median: %.2f pennies ($%.2f)\n', median(coinjar_pennies),
84 median(coinjar_pennies)/100);
85 fprintf(' Range: [%.0f, %.0f]\n\n', min(coinjar_pennies),
86 max(coinjar_pennies));
87
88 fprintf('CAR (Bernoulli p=0.88 + Negative Binomial):\n');
89 fprintf(' Mean: %.2f pennies\n', mean(car_pennies));
90 fprintf(' Std Dev: %.2f\n', std(car_pennies));
91 fprintf(' Median: %.2f\n', median(car_pennies));
92 fprintf(' % with zero: %.1f%\n', 100*sum(car_pennies==0)/n);
93 fprintf(' Range: [%d, %d]\n\n', min(car_pennies), max(car_pennies));
94
95 fprintf('DRAWER (Bernoulli p=0.95 + Poisson =53):\n');
96 fprintf(' Mean: %.2f pennies\n', mean(drawer_pennies));
97 fprintf(' Std Dev: %.2f\n', std(drawer_pennies));
98 fprintf(' Median: %.2f\n', median(drawer_pennies));
99 fprintf(' % with zero: %.1f%\n\n', 100*sum(drawer_pennies==0)/n);
100
101 % Total pennies per simulation
102 total_pennies = wallet_pennies + coinjar_pennies + car_pennies + ...
103 drawer_pennies + other_pennies;
104
105 fprintf('TOTAL PENNIES PER PERSON:\n');
106 fprintf(' Mean: %.2f pennies ($%.2f)\n', mean(total_pennies),
107 mean(total_pennies)/100);
108 fprintf(' Std Dev: %.2f\n', std(total_pennies));
109 fprintf(' Median: %.2f pennies ($%.2f)\n', median(total_pennies),
110 median(total_pennies)/100);
111 fprintf(' IQR: [%.0f, %.0f]\n\n', prctile(total_pennies, 25),
112 prctile(total_pennies, 75));
113 fprintf(' 90% CI: [%.0f, %.0f]\n\n', prctile(total_pennies, 5),
114 prctile(total_pennies, 95));
115 fprintf(' 95% CI: [%.0f, %.0f]\n\n', prctile(total_pennies, 2.5),
116 prctile(total_pennies, 97.5));
117 fprintf(' 99% CI: [%.0f, %.0f]\n\n', prctile(total_pennies, 0.5),
118 prctile(total_pennies, 99.5));
119 fprintf(' 99.9% CI: [%.0f, %.0f]\n\n', prctile(total_pennies, 0.05),
120 prctile(total_pennies, 99.95));
121
122 %% VISUALIZATIONS
123 figure('Position', [100, 100, 1400, 900]);
124
125 % Wallet
126 subplot(3, 3, 1);
127 histogram(wallet_pennies, 'BinMethod', 'integers', 'FaceColor', [0.2 0.4 0.8]);
128 grid off;
129 set(gca, 'XTickLabel', [], 'YTickLabel', []);
130 box on;
131 ax = gca;
132 ax.TickDir = 'out';
133 ax.XAxis.TickLength = [0.015 0];
134 ax.YAxis.TickLength = [0.015 0];
135
136 % Coin Jar
137 subplot(3, 3, 2);
138 histogram(coinjar_pennies, 100, 'FaceColor', [0.8 0.4 0.2]);
139 grid off;
140 set(gca, 'XTickLabel', [], 'YTickLabel', []);
141 box on;
142 ax = gca;
143 ax.TickDir = 'out';
144 ax.XAxis.TickLength = [0.015 0];
145 ax.YAxis.TickLength = [0.015 0];
146
147 % Car
148 subplot(3, 3, 3);
149 histogram(min(car_pennies, 50), 'BinMethod', 'integers', 'FaceColor', [0.2 0.8
150 0.4]);
151 grid off;
152 set(gca, 'XTickLabel', [], 'YTickLabel', []);
153 box on;
154 ax = gca;
155 ax.TickDir = 'out';
156 ax.XAxis.TickLength = [0.015 0];
157 ax.YAxis.TickLength = [0.015 0];
158
159 % Drawer
160 subplot(3, 3, 4);
161 histogram(drawer_pennies, 'BinMethod', 'integers', 'FaceColor', [0.8 0.2 0.8]);
162 grid off;
163 set(gca, 'XTickLabel', [], 'YTickLabel', []);
164 box on;
165 ax = gca;
166 ax.TickDir = 'out';
167 ax.XAxis.TickLength = [0.015 0];
168 ax.YAxis.TickLength = [0.015 0];
169
170 % Other
171 subplot(3, 3, 5);
172 histogram(other_pennies, 'BinMethod', 'integers', 'FaceColor', [0.6 0.6 0.2]);
173 grid off;
174 set(gca, 'XTickLabel', [], 'YTickLabel', []);
175 box on;
176 ax = gca;
177 ax.TickDir = 'out';
178 ax.XAxis.TickLength = [0.015 0];
179 ax.YAxis.TickLength = [0.015 0];
180
181 % Total Distribution
182 subplot(3, 3, 6);
183 histogram(total_pennies, 100, 'BinWidth', 20, 'FaceColor', [0.4 0.4 0.4]);
184 grid off;
185 set(gca, 'XTickLabel', [], 'YTickLabel', []);
186 box on;
187 ax = gca;
188 ax.TickDir = 'out';
189 ax.XAxis.TickLength = [0.015 0];
190 ax.YAxis.TickLength = [0.015 0];

```



```

187
188 % Box plot comparison
189 subplot(3, 3, [7 8 9]);
190 data_matrix = [min(coinjar_pennies,1000)];
191 boxplot(data_matrix, 'Labels', {'Jar (capped)'});
192 set(gca, 'XTickLabel', [], 'YTickLabel', []);
193 grid off;
194
195 sgtitle('Monte Carlo Simulation: 100,000 Samples per Distribution', 'FontSize',
14, 'FontWeight', 'bold');

```

```

196
197 %% EXPORT DATA (optional)
198 % Uncomment to save results to CSV
199 % results_table = table(wallet_pennies, coinjar_pennies, car_pennies, ...
200 %     drawer_pennies, other_pennies, total_pennies, ...
201 %     'VariableNames', {'Wallet', 'CoinJar', 'Car', ...
202 %     'Drawer', 'Other', 'Total'});
203 % writetable(results_table, 'penny_simulation_results.csv');

```

8.2 Appendix 2: Code for Logistical Model

./model2.py

```

1  %% =====
2  % DENSITY-DEPENDENT LOGISTIC DECAY MODEL
3
4  %  $P(d,t) = P_0(d) / [P_0(d) + (1 - P_0(d)) \exp(k(d) \cdot t)]$ 
5  %  $k(d) = a \ln(d + 1)$ 
6  %
7  % All density axes plotted as log10(density)
8  % =====
9
10 clear; close all; clc;
11
12 %% =====
13 % SECTION 1: MODEL PARAMETERS
14 %% =====
15
16 P0_func = @(d) 0.99;
17 a = 0.163;
18
19 % Density range [structures / km^2]
20 d_min = 10;
21 d_max = 50000;
22 d_values = linspace(d_min, d_max, 200);
23 log_d = log10(d_values);
24
25 % Time range [years]
26 t_max = 30;
27 t = linspace(0, t_max, 1000);
28
29 %% =====
30 % SECTION 2: MODEL COMPUTATION
31 %% =====
32
33 P = zeros(length(d_values), length(t));
34
35 for i = 1:length(d_values)
36     d = d_values(i);
37     k = a * log(d + 1);
38     P0 = P0_func(d);
39     P(i,:) = P0 ./ (P0 + (1 - P0) * exp(k * t));
40 end
41
42 k_values = a * log(d_values + 1);
43
44 %% =====
45 % SECTION 3: 2D VISUALIZATION
46 %% =====
47
48 figure('Position',[100 100 1200 500],'Color','w');
49
50 % --- Panel A: Preservation vs Time ---
51 subplot(1,2,1); hold on;
52
53 density_indices = round(linspace(1,length(d_values),6));
54 colors = jet(length(density_indices));
55
56 for idx = 1:length(density_indices)
57     plot(t, P(density_indices(idx),:), ...
58          'Color', colors(idx,:), 'LineWidth', 2);
59 end
60
61 xlabel('Time (years)');
62 ylabel('Preservation Probability, P(d,t)');
63 title('Preservation Decay Over Time');
64 ylim([0 1]);
65 grid on;
66 box on;
67 hold off;
68
69 % --- Panel B: Decay Rate vs log10(Density) ---
70 subplot(1,2,2);
71
72 plot(log_d, k_values, 'LineWidth', 2.5, 'Color',[0.85 0.33 0.10]);
73
74 xlabel('log_{10}(Density) (structures km^{-2})');
75 ylabel('Decay Rate, k(d)');
76 title('Density-Dependent Decay Rate');
77 set(gca, 'YDir', 'reverse');
78 grid on;
79 box on;
80
81 %% =====
82 % SECTION 4: 3D PRESERVATION SURFACE
83 %% =====
84
85 figure('Position',[100 100 900 700],'Color','w');
86
87 [D_mesh, T_mesh] = meshgrid(log_d, t);
88 P_mesh = P';
89
90 surf(D_mesh, T_mesh, P_mesh, 'EdgeColor','none');
91
92 xlabel('log_{10}(Density)');

```

```

93 ylabel('Time (years)');
94 zlabel('Preservation Probability');
95 title('Preservation Surface P(d,t)');
96
97 colormap(jet);
98 colorbar;
99 zlim([0 1]);
100 view(45,30);
101 grid on;
102 box on;
103
104 %% =====
105 % SECTION 5: HEATMAP (log10 density)
106 %% =====
107
108 figure('Position',[100 100 900 600],'Color','w');
109
110 imagesc(log_d, t, P_mesh);
111 set(gca, 'YDir', 'normal');
112
113 xlabel('log_{10}(Density)');
114 ylabel('Time (years)');
115 title('Preservation Probability Heatmap');
116
117 colormap(jet);
118 colorbar;
119 grid on;
120 box on;
121
122 hold on;
123 contour(log_d, t, P_mesh, ...
124         [0.9 0.7 0.5 0.3], 'k', 'LineWidth', 1.2);
125 hold off;
126
127 %% =====
128 % SECTION 6: KEY METRICS (log10 density)
129 %% =====
130
131 figure('Position',[100 100 1200 500],'Color','w');
132
133 % --- Panel A: Time to 50% Preservation ---
134 subplot(1,2,1);
135
136 P_threshold = 0.5;
137 t_half = NaN(size(d_values));
138
139 for i = 1:length(d_values)
140     idx = find(P(i,:) <= P_threshold, 1);
141     if ~isempty(idx)
142         t_half(i) = t(idx);
143     end
144 end
145
146 log_d_interp = linspace(log10(d_min), log10(d_max), 1000);
147 t_half_interp = interp1(log_d, t_half, log_d_interp, 'pchip');
148
149 plot(log_d_interp, t_half_interp, ...
150      'LineWidth', 2.5, 'Color', [0.47 0.67 0.19]);
151
152 xlabel('log_{10}(Density)');
153 ylabel('Time to 50% Preservation (years)');
154 title('Preservation Half-Life vs Density');
155 grid on;
156 box on;
157
158 % --- Panel B: Decay Rate Reference ---
159 subplot(1,2,2);
160
161 plot(log_d, k_values, ...
162      'LineWidth', 2.5, 'Color', [0.49 0.18 0.56]);
163
164 xlabel('log_{10}(Density)');
165 ylabel('Decay Rate, k(d)');
166 title('Decay Rate as Function of Density');
167 set(gca, 'YDir', 'reverse');
168 grid on;
169 box on;
170
171 %% =====
172 % SECTION 7: SUMMARY
173 %% =====
174
175 fprintf('\n===== \n');
176 fprintf('MODEL SUMMARY STATISTICS \n');
177 fprintf('===== \n');
178 fprintf('Decay scaling parameter a: %.4f \n', a);
179 fprintf('Density range: %.0f - %.0f structures/km^2 \n', d_min, d_max);
180 fprintf('Time horizon: 0 - %.0f years \n', t_max);
181 fprintf('Decay-rate range: %.4f - %.4f \n', min(k_values), max(k_values));
182 fprintf('Initial preservation P0: %.2f \n', P0_func(d_min));
183 fprintf('===== \n \n');

```

8.3 Appendix 3: Code for Rounding Strategy

`./model3.m`

```

1  % Visualization of J(theta, p)
2  % J(, p) = C_0 * (r*/2 + 0.02*p) * (1 - e^(-r**T)) / (2**||)
3
4  clear; clc;
5
6  % Parameters
7  C_0 = 240 * 0.25; % 60
8  r = 0.05;
9  lambda = 0.8; % Decay rate
10 T = 120;
11
12 % Define ranges
13 theta_range = linspace(-1, 1, 100);
14 p_range = linspace(0.1, 5, 100);
15
16 % Create meshgrid
17 [THETA, P] = meshgrid(theta_range, p_range);
18
19 % Calculate J
20 THETA_safe = THETA;
21 THETA_safe(THETA == 0) = eps; % Avoid division by zero
22
23 exponent = -lambda .* r .* THETA_safe .* T;
24 J = -C_0 .* ((r .* THETA_safe / 2) + 0.02 .* P) .* (1 - exp(exponent)) ./
    ((THETA_safe) .* lambda .* r);
25
26 %% Find optimal theta for each p
27 optimal_theta = zeros(size(p_range));
28 for i = 1:length(p_range)
29     [~, max_idx] = max(J(i, :));
30     optimal_theta(i) = theta_range(max_idx);
31 end
32
33 %% VISUALIZATIONS
34 figure('Position', [100, 100, 900, 700]);
35
36 % Heat map / Image plot with log scale coloring
37 % Use sign(J).*log10(abs(J)+1) to handle negative values
38 J_log = J;
39 imagesc(theta_range, p_range, J_log);
40 set(gca, 'YDir', 'normal'); % Flip y-axis to normal orientation
41 hold on;
42
43 % Draw optimal theta line
44 plot(optimal_theta, p_range, 'w-', 'LineWidth', 3);
45 plot(optimal_theta, p_range, 'k--', 'LineWidth', 1.5);
46
47 hold off;
48 xlabel(' (theta)', 'FontWeight', 'bold', 'FontSize', 12);
49 ylabel('p (price)', 'FontWeight', 'bold', 'FontSize', 12);
50
51 title(sprintf('J(, p) Heat Map with Optimal \nC_0 = 240.25 = %.0f, r = %.2f, =
    %.1f, T = %d', ...
52         C_0, r, lambda, T), 'FontSize', 13, 'FontWeight', 'bold');
53
54 % Colorbar with blue to red colormap
55 cb = colorbar;
56 cb.Label.String = 'sign(J) * log(|J| + 1)';
57 cb.Label.FontSize = 12;
58 cb.Label.FontWeight = 'bold';
59 colormap(jet); % Blue to red through cyan-green-yellow
60
61 axis tight;
62 grid off;
63
64 %% STATISTICS
65 fprintf('=== FUNCTION STATISTICS ===\n');
66 fprintf('Parameters:\n');
67 fprintf(' C_0 = 240 * 0.25 = %.0f\n', C_0);
68 fprintf(' r = %.2f\n', r);
69 fprintf(' lambda = %.1f\n', lambda);
70 fprintf(' T = %d\n', T);
71
72 fprintf('J(, p) Statistics:\n');
73 fprintf(' Min J: %.4f\n', min(J(:)));
74 fprintf(' Max J: %.4f\n', max(J(:)));
75 fprintf(' Mean J: %.4f\n', mean(J(:)));
76 fprintf(' Std J: %.4f\n', std(J(:)));
77
78 % Sample values
79 fprintf('Sample Values:\n');
80 test_points = [0.5, 2; -0.5, 2; 0.8, 5; -0.8, 8];
81 for i = 1:size(test_points, 1)
82     theta_test = test_points(i, 1);
83     p_test = test_points(i, 2);
84
85     if theta_test == 0
86         J_test = 0;
87     else
88         exp_test = -lambda * r * theta_test * T;
89         J_test = C_0 * ((r * theta_test / 2) + 0.02 * p_test) * ...
90             (1 - exp(exp_test)) / (abs(theta_test) * lambda * 2);
91     end
92
93     fprintf(' J(=%.1f, p=%.1f) = %.4f\n', theta_test, p_test, J_test);
94 end
95
96 fprintf('\n=== OPTIMAL THETA VALUES ===\n');
97 fprintf('p = %.2f: * = %.4f\n', p_range(1), optimal_theta(1));
98 fprintf('p = %.2f: * = %.4f\n', p_range(50), optimal_theta(50));
99 fprintf('p = %.2f: * = %.4f\n', p_range(end), optimal_theta(end));

```

9 AI REPORT

AI was not used in the making of this project.